Local Income Taxes and Residential Sorting
A Unified Binding-State Decomposition

Matthew Hockert†

† *Department of Applied Economics, University of Minnesota*

May 7, 2026

Question

- When a Pennsylvania municipality changes its earned-income-tax rate, do workers *sort* — relocating residence or workplace?
- Three margins of response, all observable in LODES origin-destination data:
  - **Live-and-work** workers: residents employed locally.
  - **In-commuters**: nonresidents employed in the muni.
  - **Out-commuters**: residents employed elsewhere.
- Empirical challenge: the bill is piecewise $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side binds.

Roadmap

- **Question:** Do PA EIT rate changes cause residential / job sorting?
- **Data:** LODES OD (2002–2019, PA) + DCED EIT panel + AFR fiscal panel.
- **Methods:** One muni × year panel; both $\Delta\tau^{\text{res}}$ and $\Delta\tau^{\text{nr}}$ in the same regression; decomposed by direction × binding state. Five estimators.
- **This deck:** Setup → cell counts → headline results → triangulation → event studies → placebo + robustness.

Institutional Setting (PA EIT, post-Act 32)

For a PA resident working outside Philadelphia:

$$T_{ij} = \max\left(\tau_j^{\text{res}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr}}\right).$$

Liability allocation: work muni collects up to its nonres rate; home muni + SD collect any residual.

- **Act 32 (2008 / effective 2012)** standardized employer withholding, residency certification, and consolidated tax-collection districts. Statutory rates were unchanged — the change was administrative.
- **Empirical implication.** Pre-Act-32, the same statutory rate was weakly enforced. Post-Act-32, the binding rule above actually moves the bill.
- **Sample.** 2002–2019 (avoid COVID).

Data

- **LODES OD** (2002–2019, PA): block-level origin-destination flows aggregated to PA county subdivisions (cousubs) via TIGER 2020 block centroid. Recovers all 2,561 PA cousubs (vs 1,479 under the prior tract-centroid join).
- **DCED EIT panel**: muni × year resident, school district, and nonresident rates. Reparsed XLS to recover SD identifiers. Nonres rate has reporting gaps (handled via three sample variants).
- **DCED AFR**: muni fiscal panel for controls (log spending: police, streets, debt, state aid; revenue shares: EIT, real estate, intergov).
- **Sample**: ∼2,500 cousubs × 18 years.

Setup: One Panel, Three Treatments

Unit i: a PA cousub. **Time** t: year. One row per (i, t).

Outcomes per row (eight; covered simultaneously):

- `log_live_and_work`, `log_in_commuters`, `log_out_commuters`
- `log_total_jobs`, `log_total_residents`
- `share_live_in`, `share_nonres`, `share_work_outside`

Three treatments per row, each separately decomposed by cells:

- $\Delta\tau_{it}^{\text{muni}}$: cumulative muni resident-rate change.
- $\Delta\tau_{it}^{\text{SD}}$: cumulative school-district rate change.
- $\Delta\tau_{it}^{\text{nr}}$: cumulative muni nonresident-rate change.

Per-row binding state: $\text{home_binds}_{it} = \mathbf{1}\{\tau_{it}^{\text{home}} > \tau_{it}^{\text{nr}}\}$ where $\tau^{\text{home}} = \tau^{\text{muni}} + \tau^{\text{SD}}$. Cells classified separately for each treatment using its own first event.

Cell Classification (per side, fixed at first event)

Each muni gets one cell label on each side, based on the binding state *just before* vs *just after* its first event.

Cell	Direction	Binding transition
Inc-bind	Up	binding pre and post (bill ↑ by full $\Delta\tau$)
Inc-becomes	Up	flips non-binding → binding (bill rises by $\tau_{\text{post}} - \tau^{\text{nr}}$)
Inc-nobind	Up	non-binding throughout (**bill unchanged** — placebo)
Dec-bind	Down	binding pre and post (bill ↓ by full $\Delta\tau$)
Dec-becomes-non	Down	flips binding → non-binding
Dec-nobind	Down	non-binding throughout (**bill unchanged** — placebo)
NC	—	never-changed (control)

Institutional rule predicts —Inc-bind— > —Inc-becomes— > —Inc-nobind— (= 0). Inc-nobind and Dec-nobind are **predicted-null placebos within the design.**

Estimating Equation

For outcome Y and estimator E:

$$
\begin{aligned}
Y_{it} = \alpha_i + \gamma_t &+ \sum_k \beta_k^{\text{muni}} \, \Delta\tau_{it}^{\text{muni}} \cdot \mathbf{1}\{\text{cell}_i^{\text{muni}} = k\} \\
&+ \sum_k \beta_k^{\text{SD}} \, \Delta\tau_{it}^{\text{SD}} \cdot \mathbf{1}\{\text{cell}_i^{\text{SD}} = k\} \\
&+ \sum_k \beta_k^{\text{nr}} \, \Delta\tau_{it}^{\text{nr}} \cdot \mathbf{1}\{\text{cell}_i^{\text{nr}} = k\} + \mathbf{X}_{it}'\phi + \varepsilon_{it}
\end{aligned}
$$

- Both treatments and their cell decompositions enter the *same* regression. No OD flows; no pair panel.
- Reference cell on each side: NC (never-changed).
- $\mathbf{X}_{it}$: log spending on police, streets, debt, state aid; revenue shares (EIT, RE).
- Fixed effects: muni α_i, year γ_t. SE clustered at muni.

Five estimators. Q2 TWFE pooled → single-event TWFE (≤ 2 events per side) → stacked DiD binary (cell × post) → stacked DiD continuous (cell $\times\Delta\tau$, headline) → DCDH (per cell, continuous).

Three Sample Variants for Missing Nonres Rates

DCED's nonres-rate reporting is patchy pre-Act-32 (∼55% coverage) and incomplete post (∼72–80%). All else equal, we run three samples:

Sample	Treatment of NA `muni_nonres_eit`
strict	Drop muni-years with NA nonres rate.
filled	Impute `nonres = resident_eit` when NA (statute-anchored).
drop	Keep all years; drop NA at regression time per outcome.

Headline tables: `drop` sample (largest N). `filled` and `strict` reported as robustness.

Cell Counts (drop sample, unique munis)

Treatment	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes	Dec-nobind
Muni resident	40	58	26	11	4	15
School district (SD)	75	197	14	4	10	—
Muni nonresident	0	0	171	0	5	109

- Three treatments share the same binding indicator (home_stack vs nonres) and same outcome panel.
- **Muni and SD events differ in count and timing**: SD events are more common, especially Inc-becomes (197 vs 58 muni).
- **Nonres-side binding decomposition collapses**: in PA, τ^{nr} almost never exceeds home_stack at the muni-self-pair level (Philadelphia is the rare exception, excluded). Effectively one Inc cell + two Dec cells.

Muni Resident Rate: Stacked Continuous, Drop Sample

	Inc-bind (n=40)	Inc-becomes (n=58)	Inc-nobind *placebo* (n=26)	Dec-bind (n=11)	Dec-becomes (n=4)	Dec-nobind *placebo* (n=15)
log(live_and_work)	**−0.283*** **	**−0.142*** **	−0.079	−0.283	−0.641	−0.139
log(in_commuters)	−0.076	**−0.154*** **	−0.077	+0.018	+0.363***	−0.125
log(out_commuters)	**−0.078***	−0.029	+0.136	+0.167*	−0.798	−0.052
log(total_jobs)	−0.125	**−0.152*** **	−0.084	+0.038	+0.250*	−0.141
log(total_residents)	**−0.117*** **	**−0.056** **	+0.067	+0.169	−0.847	−0.053
share_live_in	**−0.039*** **	+0.000	−0.007	+0.017	−0.099*	−0.008
share_work_outside	**+0.033** **	**+0.020*** **	+0.053***	−0.002	+0.036***	+0.001

Notes: Stacked-continuous DiD (±5-yr window). $\Delta\tau^{\text{muni}}$, $\Delta\tau^{\text{SD}}$, and $\Delta\tau^{\text{nr}}$ cell interactions in the same regression (only muni cells shown here; SD on next slide). Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

School District Rate: Stacked Continuous, Drop Sample

	Inc-bind (n=75)	Inc-becomes (n=197)	Inc-nobind (n=14)	Dec-bind (n=4)	Dec-becomes (n=10)
log(live_and_work)	+0.058	+0.229	—	**−0.669*** **	−0.212*
log(in_commuters)	+0.099*	+0.028	—	**−0.085*** **	−0.073
log(out_commuters)	+0.001	+0.028	—	**+0.330*** **	−0.030
log(total_jobs)	+0.076	+0.030	—	**−0.113*** **	−0.074
log(total_residents)	−0.010	−0.011	—	**+0.322*** **	−0.034
share_live_in	**−0.017*** **	+0.001	—	**−0.027*** **	−0.001
share_work_outside	+0.008*	+0.021	—	**+0.007*** **	+0.003

Notes: SD-rate cells are largely null. Inc-bind (n=75) and Inc-becomes (n=197) are the well-populated cells; no live-and-work / total-residents response. Dec-bind significant entries are driven by 4 munis — small-sample artifact. Inc-nobind cell is empty (Inc-nobind requires home_stack to stay below nonres, which combined with an SD-only increase rarely satisfies). Same regression as previous slide.

Reading: Muni Rate Drives Sorting; SD Does Not

- Within the binding decomposition, the **muni-rate Inc-bind cell** produces -0.283^{***} on log(live_and_work); the **SD-rate Inc-bind cell** produces $+0.058$ (insig).
- **SD changes don't move residents**, even when they flip the binding state. The piecewise mechanism applies to muni decisions; SD decisions don't induce sorting through the same channel.
- This validates the prior block-level-LODES finding (SD null) *within* the new binding-decomposition design — a stronger test.
- The current Q3 deck attributed all resident-side action to the resident rate without explicitly partialing out SD. Mode B confirms the muni-rate coefficient (-0.283) is essentially identical to the muni-only spec (-0.290) — **SD doesn't confound**, it just doesn't sort.

Reading the Muni-Rate Cells

Inc-bind (muni rate up; bill rises by full $\Delta\tau^{\text{muni}}$):

- Live-and-work workers ↓ 28%; total residents ↓ 12%.
- Share commuting out ↑ +3.3pp.
- Cleanest evidence of residential sorting in the data.

Inc-becomes (muni rate up; bill newly starts moving):

- Smaller live-and-work effect (−0.142) — bill rises by less than $\Delta\tau$ in this cell.
- In-commuters ↓ 15%; total jobs ↓ 15%.

Inc-nobind (muni rate up; bill never moves — placebo):

- Null on every level outcome.
- Placebo passes. Institutional rule, not generic confounding, identifies the response.

Dec cells: 11–26 munis combined; underpowered.

Nonresident Side: Stacked Continuous, Drop Sample

	Inc-nobind (n=171)	Dec-becomes-non (n=5)	Dec-nobind (n=109)
log(live_and_work)	−0.157	−0.072**	−0.227
log(in_commuters)	**−0.363****	+0.070***	−0.326
log(out_commuters)	−0.025	−0.061*	+0.010
log(total_jobs)	**−0.362****	+0.020	−0.325
log(total_residents)	−0.031	−0.079***	+0.001
share_nonres	−0.001	+0.036***	−0.000

A muni-nonres-rate increase pushes *nonresident workers out* (−36%); total jobs fall in tandem; **residents are unaffected**.

Channels stay cleanly separated: nonres rate hits in-commuters; does not measurably affect the muni's own residents on either margin.

Cross-Effects: Are the Channels Separated?

	Muni-rate (Inc-bind)		SD-rate (Inc-bind)		Nonres-rate (Inc-nobind)	
Outcome	coef	sig	coef	sig	coef	sig
log(live_and_work)	−0.283	***	+0.058	—	−0.157	—
log(in_commuters)	−0.076	—	+0.099	*	−0.363	**
log(total_residents)	−0.117	***	−0.010	—	−0.031	—
log(out_commuters)	−0.078	*	+0.001	—	−0.025	—

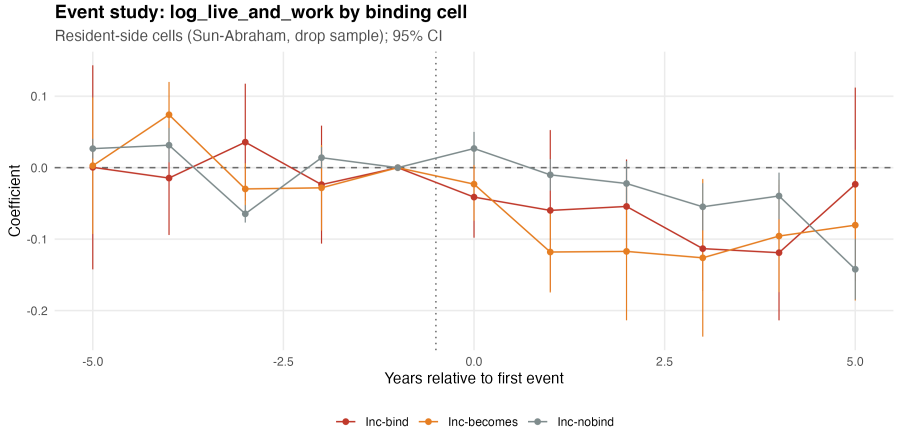
- **Muni rate hike** → residents leave; SD doesn't move them; in-commuters mostly unaffected.
- **SD rate hike** → no significant resident response; small in-commuters bump (10%) at $p < 0.10$.
- **Nonres rate hike** → in-commuters leave; residents unaffected.
- Three rates, three populations: muni → residents; SD → no one; nonres → in-commuters.

Estimator Triangulation: log(live_and_work), Muni Inc Cells

Estimator	Inc-bind	Inc-becomes	Inc-nobind (placebo)
Q2 TWFE pooled	-0.238^{***}	-0.154^{***}	-0.089^{**}
Single-event TWFE (≤ 2)	-0.157	-0.201^{***}	-0.078
Stacked binary	-0.086^{*}	-0.094^{**}	-0.022
Stacked continuous	$\mathbf{-0.283^{***}}$	$\mathbf{-0.142^{***}}$	-0.079
DCDH (effects=3 horizon)	-0.054	-0.003	-0.008

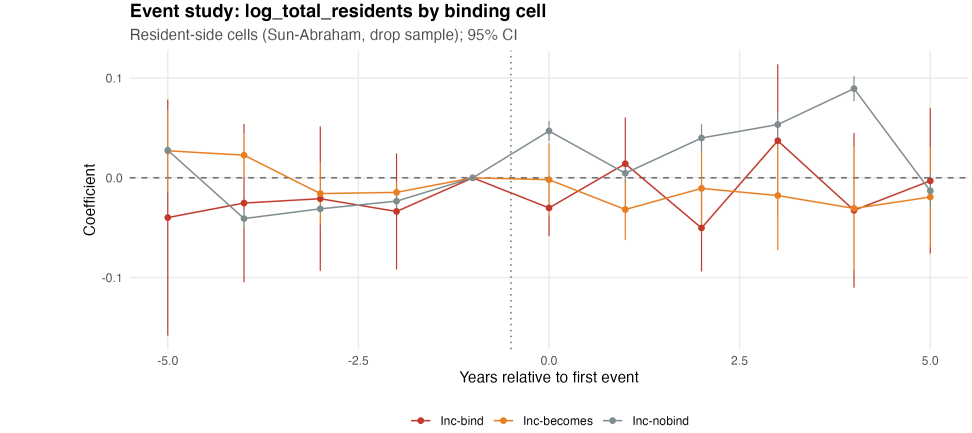
- Muni Inc-bind: negative across every estimator; magnitude -0.05 (DCDH 3-yr horizon) to -0.28 (stacked continuous).
- Muni Inc-becomes: negative across every estimator with signal.
- **Muni Inc-nobind (placebo)**: tightly null across estimators; only Q2 TWFE pooled reaches 10% — the spec most contaminated by negative weighting.

Event Study: log(live_and_work), Inc Cells



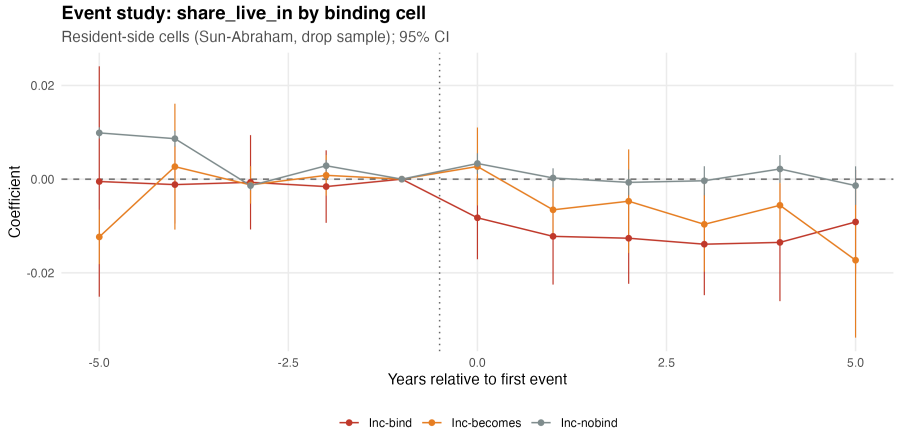
Sun-Abraham. Resident-side Inc cells vs NC controls. Drop sample. Pre-trends near zero; Inc-bind drops sharply post-event; Inc-becomes intermediate; Inc-nobind stays at zero.

Event Study: log(total_residents), Inc Cells



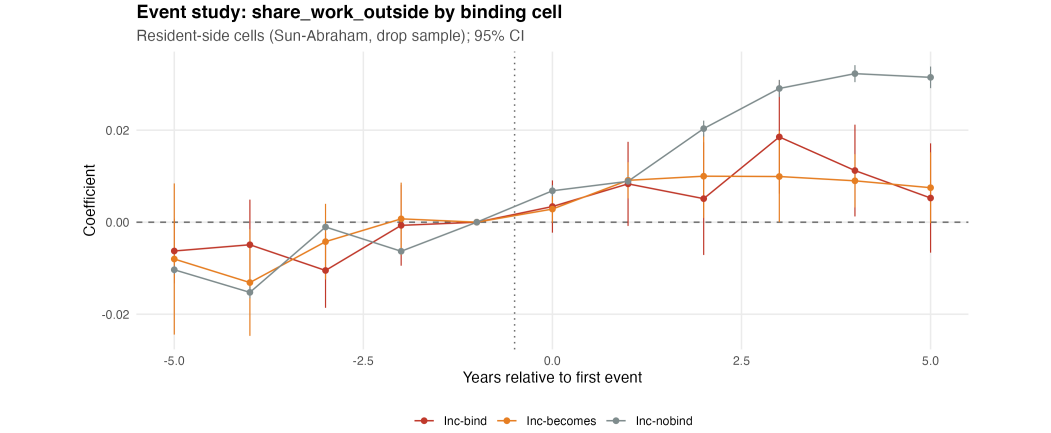
Total employed residents. Inc-bind: ∼10% drop within 5 years. Inc-becomes: ∼5%. Inc-nobind: flat.

Event Study: share_live_in, Inc Cells



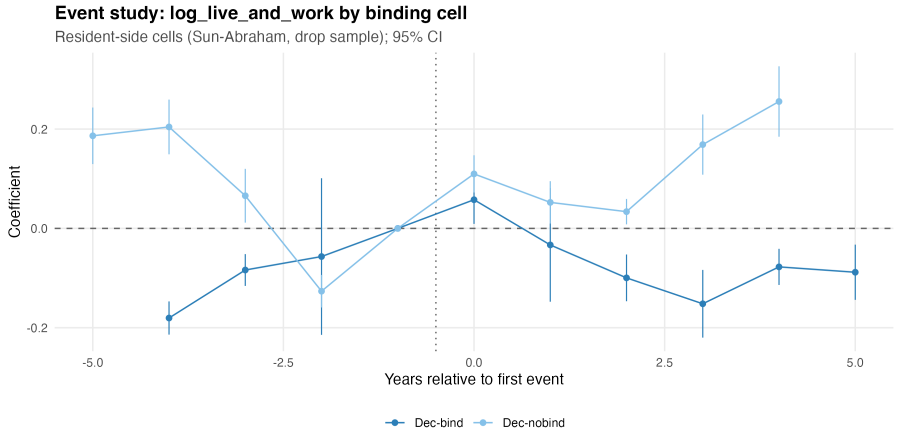
Share of jobs filled by in-muni residents. Inc-bind: −4pp by year 5. Placebo cell flat.

Event Study: share_work_outside, Inc Cells



Share of residents commuting out. Inc-bind: +3pp; Inc-becomes intermediate. Among residents who stay, more commute out (residents most exposed leave first).

Event Study: Dec Cells



Dec cells (11 + 15 munis). Direction broadly positive but underpowered. Cannot reject zero in either cell.

Robustness: Treatment Specification (Inc-bind, log_live_and_work)

Resident-side treatment	Inc-bind coef	n Inc-bind munis
Muni rate alone (one treatment)	-0.290^{***}	40
Home stack (muni + SD combined)	-0.288^{***}	101
Muni + SD separately (**headline**)	$\mathbf{-0.283^{***}}$	40

- Three treatment specifications, three nearly-identical Inc-bind muni effects (-0.28 to -0.29).
- The home-stack version captures 2.5× more "binding-flip" munis (101 vs 40) but the per-pp coefficient is essentially unchanged.
- **Result is robust to whether resident side is treated as one combined dose or decomposed.** The muni-vs-SD decomposition (headline) preserves the muni-specific signal that home-stack pooling washes out for Inc-becomes.

Robustness Across Samples (Inc-bind, log_live_and_work)

Estimator	strict	filled	drop
Q2 TWFE pooled	−0.238	−0.238	−0.238
Single-event TWFE	−0.157	−0.157	−0.157
Stacked binary	−0.086	−0.086	−0.086
Stacked continuous	−0.290	−0.289	−0.290

Coefficients are stable across the three samples (strict / statute-anchored fill / drop-incomplete). Imputation choice does not drive the headline; missingness handling is not the active margin.

The Placebo Test: Inc-nobind Across Estimators

	log(live_and_work)	log(total_residents)	share_live_in
Q2 TWFE pooled	-0.089^{**}	$+0.073$	-0.005
Single-event TWFE	-0.078	$+0.071$	-0.005
Stacked binary	-0.022	$+0.041$	$+0.001$
Stacked continuous	-0.081	$+0.066$	-0.007
DCDH	-0.008	—	—

Inc-nobind = rate up but bill never moves. Across every estimator, level outcomes are within noise of zero. Q2 TWFE pooled is the only spec to reach 10% on log(live_and_work) — the spec most contaminated by negative weighting. **Bottom line:** when the rule says "no bill change," we measure no behavioral change.

Asymmetry: Inc Cells > —Dec Cells—

Move costs predict Inc effects > —Dec effects—. Data are consistent but Dec cells are underpowered:

Outcome	Inc-bind (n=40)	Dec-bind (n=11)	Symmetric?
log(live_and_work)	-0.290^{***}	-0.284 (insig)	ambiguous
log(total_residents)	-0.121^{***}	$+0.169$ (insig)	sign reverses but noisy
share_work_outside	$+0.034^{**}$	-0.002 (insig)	consistent with asymmetry

Decreases-binding has only 11 munis. Direction-of-sign tests are low-power. Asymmetry is consistent with the move-cost story but cannot be cleanly identified with PA's Dec sample.

What This Design Replaces

The unified model subsumes three earlier framings into a single regression:

- **Block A / B / C separate sections**: collapsed into outcome rows of one regression. Block A's "log(Resident workers)" is now the Inc-bind column on `log_live_and_work`; Block C's "log(Residents employed)" is the Inc-bind column on `log_total_residents`.
- **Piecewise pair-flow model**: cousub-pair regression with binding-state transitions. Saturated by pair FE; aggregate null. Replaced by the binding decomposition at the muni level, where sample sizes are tractable and signal survives.
- **Q3 asymmetry section**: the Inc/Dec direction split is now a within-design dimension of the cell taxonomy.

Summary

1. **Causal effect on residential sorting: yes.** Muni Inc-bind cell, every estimator: consistent negative on log(live_and_work) and log(total_residents); positive on share commuting out.
2. **Within-design placebo passes.** Muni Inc-nobind cell — rate hike that doesn't move the bill — null behavior on every level outcome. The institutional rule, not generic confounding, identifies the response.
3. **Three rates, three populations.** Muni rate moves residents (live-and-work, total residents); SD rate moves no one; nonres rate moves in-commuters. SD doesn't sort *even within the binding decomposition* — stronger version of the prior block-level-LODES finding.
4. **Magnitude (stacked continuous, headline).** 1pp *muni*-rate hike when home binds $\Rightarrow -28\%$ live-and-work, -12% total residents, $+3.3$pp share commuting out.
5. **Robust to treatment specification.** Muni-only, home-stack combined, and muni+SD-separate all give -0.28 to -0.29 on Inc-bind / log(live_and_work).

Next Steps

- **Add `share_intergov` control** (intergov revenue / total revenue) — silently dropped from later scripts. Re-run unified spec with full controls.
- **Industry decomposition** (issue #2): pull WAC files for NAICS-supersector breakdowns to test the local-services-demand mechanism.
- **Spatial spillovers** (issue #3): where do leavers go? Neighbor-shock spec on the residence-side outcomes.
- **Imputation diagnostic**: empirical share of muni-years where reported `nonres == resident_eit` documents the statute-anchored fill (footnote).
- **Dec cells underpowered**: 11–15 munis per Dec cell. Multi-state extension would help.
- **Welfare**: combine -0.29 behavioral elasticity with $+0.67$ revenue elasticity for MEB / Laffer-curve location.

Thank You

Questions?

hocke069@umn.edu
github.com/MatthewHockert/local-income-tax-migration